

Krakow Tram Network Simulation

A Discrete Event Simulation of Public Transport Operations

Course: Discrete System Simulations

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Abstract

This report presents the development and validation of a discrete event simulation model for the Krakow tram network. The simulation accurately models tram movements, passenger arrivals, and boarding/alighting processes using real-world schedule data and calibrated demand profiles. By integrating data from multiple sources including GTFS feeds, geospatial datasets, and official transport statistics, we created a high-fidelity model that closely matches observed passenger volumes. The simulation enables analysis of system efficiency, peak-hour behavior, and the impact of parameter modifications on network performance. Key innovations include weighted stop popularity models, gravity-based destination selection, and dynamic demand curves based on comprehensive transport studies.

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1 Introduction

1.1 Topic Selection and Motivation

Public transportation systems play a crucial role in urban mobility, especially in densely populated European cities. Krakow, Poland's second-largest city with a metropolitan population exceeding 800 thousand, relies heavily on its extensive tram network. The network consists of over 20 active lines serving hundreds of stops, providing passengers with approximately 621,000 journeys daily according to the 2023 Comprehensive Transport Study (KBR 2023).

Understanding the dynamics of such complex systems requires sophisticated modeling techniques. Discrete Event Simulation (DES) provides an ideal framework for analyzing public transport operations because:

- **Stochastic Nature:** Passenger arrivals follow probabilistic patterns that vary throughout the day
- **Resource Constraints:** Trams have limited capacity
- **Time-Dependent Behavior:** Demand peaks during morning and afternoon rush hours create bottlenecks
- **Network Interactions:** Multiple lines share stops and tracks, creating interdependencies

Our simulation project addresses these challenges by creating a validated model that can serve as a decision-support tool for transport planning, capacity analysis, and service optimization.

1.2 Importance of Simulation in Public Transport Planning

Simulation models are indispensable tools in modern transportation planning for several reasons:

1. **Cost-Effective Testing:** Evaluating schedule changes, new lines, or infrastructure modifications without disrupting real operations
2. **Capacity Planning:** Understanding how system performance degrades under increasing demand
3. **Pandemic Response:** Modeling social distancing requirements and reduced capacity scenarios
4. **Energy Efficiency:** Optimizing tram schedules to reduce empty runs and improve utilization
5. **Accessibility Analysis:** Identifying underserved areas and evaluating service equity

The accuracy of our model, validated against real-world statistics, demonstrates that discrete event simulation can reliably represent complex transport networks when properly calibrated with empirical data.

2 Model Description

2.1 Discrete Event Simulation Framework

Our simulation is built on the principles of Discrete Event Simulation (DES), where the system state changes only at discrete points in time when events occur. The simulation uses SimPy, a process-based discrete-event simulation framework for Python.

2.1.1 Core Entities

The simulation model consists of three primary entity types:

1. **Trams:** Represented as `TramBlock` objects containing scheduled trips with stop times
2. **Passengers:** Individual entities with origin stops, destination stops, and state (WAITING, ON_TRAM, ALIGHTED)
3. **Stops:** Physical locations where trams arrive, passengers board/alight, and waiting queues form

2.1.2 Event Types

The simulation processes the following event categories:

- **Tram Arrival Events:** Scheduled times when a tram block arrives at a stop according to GTFS data
- **Passenger Generation Events:** Continuous stochastic process generating passengers at stops based on demand curves
- **Boarding Events:** Passengers enter trams when conditions are met (capacity, destination reachability)
- **Alighting Events:** Passengers exit trams upon reaching their destination stops

2.2 Passenger Arrival Model

The passenger arrival process is the most critical stochastic component of our simulation. We model arrivals using a multi-layered approach:

2.2.1 Global Demand Curve

The system-wide demand rate $\lambda(t)$ at time t (in minutes from midnight) is modeled as:

$$\lambda(t) = \lambda_{base} + \sum_{i=1}^n h_i \cdot \exp\left(-\frac{(t - t_i)^2}{2w_i^2}\right) \quad (1)$$

where:

- λ_{base} = base demand rate (constant background traffic)

- n = number of demand peaks
- h_i = height (intensity) of peak i
- t_i = time center of peak i (minutes from midnight)
- w_i = width parameter of peak i (standard deviation of Gaussian)

Based on KBR 2023 data analysis, we calibrated three peaks:

- **Morning Peak:** $t_1 = 450$ minutes (7:30 AM), $w_1 = 90$ minutes, $h_1 = 700$ passengers/minute
- **Midday Peak:** $t_2 = 810$ minutes (1:30 PM), $w_2 = 150$ minutes, $h_2 = 420$ passengers/minute
- **Evening Peak:** $t_3 = 1020$ minutes (5:00 PM), $w_3 = 120$ minutes, $h_3 = 700$ passengers/minute

2.2.2 Stop-Level Distribution

The global demand is distributed among stops based on weighted popularity and accessibility. Each stop s receives a generation weight w_s calculated as:

$$w_s = w_{base}(s) \cdot a_s \quad (2)$$

where:

- $w_{base}(s)$ = configurable base weight for stop s (user-defined popularity weight, default: 1.0)
- a_s = gravity-based accessibility score for stop s , computed as:

$$a_s = \sum_{d \in D_s} \frac{w_{base}(d)}{d_{sd} + 800} \quad (3)$$

where D_s is the set of all destinations reachable from stop s , $w_{base}(d)$ is the base weight of destination d , and d_{sd} is the minimum travel distance (in meters) from stop s to destination d along available tram routes. The constant 800 meters acts as a distance offset to prevent division by zero and model real-world walking/transit preferences.

The accessibility score a_s captures how well-connected a stop is to attractive destinations, with higher scores for stops that can reach many popular destinations with short travel distances. This gravity-based approach reflects that stops with better connectivity and proximity to attractive destinations generate more passenger demand.

The arrival rate at stop s at time t is:

$$\lambda_s(t) = \lambda(t) \cdot \frac{w_s}{\sum_{s'} w_{s'}} \quad (4)$$

where $\sum_{s'} w_{s'}$ is the total system weight across all boarding-capable stops.

2.2.3 Passenger Generation Process

For each stop s and time interval $[t, t + \Delta t]$, the number of arriving passengers is approximated using a Poisson-like process:

$$N_s(t, t + \Delta t) \approx \text{Poisson}(\lambda_s(t) \cdot \Delta t) \quad (5)$$

The implementation uses an approximation scheme:

- When $\lambda_s(t) \cdot \Delta t \geq 1$: The integer part is deterministically generated, plus an additional passenger with probability equal to the fractional part.
- When $\lambda_s(t) \cdot \Delta t < 1$: A single passenger is generated with probability $\lambda_s(t) \cdot \Delta t$ (Bernoulli trial).

Each generated passenger is assigned:

- **Origin stop**: The stop where generation occurred
- **Destination stop**: Selected using a gravity model (see Section 2.3), which considers distance and destination popularity
- **Target line**: A randomly selected line among those serving both the origin and destination stops
- **Arrival time**: Set to t (the start of the interval $[t, t + \Delta t]$)

2.3 Destination Selection: Gravity Model

Passenger destinations are not uniformly distributed. We employ a gravity-based model where the probability of selecting destination d from origin o depends on:

1. **Distance**: Shorter trips are more likely (inverse distance relationship with offset)
2. **Stop Popularity**: More popular stops attract more passengers (proportional to base weight w_d)
3. **Reachability**: Destination must be reachable via available tram lines (pre-computed reachability map)
4. **Service Availability**: Future service must exist within a reasonable time window (maximum wait: 30 minutes, maximum trip duration: 120 minutes)

The selection weight for destination d from origin o is:

$$P(d|o) \propto \frac{w_d}{d_{od} + 800} \quad (6)$$

where:

- w_d = base popularity weight of destination d (configurable, default: 1.0)

- d_{od} = minimum network distance between origin o and destination d , measured in meters along tram route shapes (computed as the shortest path distance across all routes serving the O-D pair, with a minimum of 50 meters)
- The constant 800 meters acts as a distance offset to prevent division by zero and model real-world transit preferences (similar to a softening parameter in inverse distance weighting)

The distance d_{od} is computed during initialization by:

1. Iterating through all tram trips in the schedule
2. For each trip, calculating the distance from origin o to destination d as the difference in cumulative shape distances along the route
3. Storing the minimum distance across all routes that serve the O-D pair

Destinations are selected from a probability distribution normalized over all reachable stops. After destination selection, a target line is randomly chosen from the set of lines that serve the origin-destination pair. The number of lines serving an O-D pair is tracked but does not influence the selection probability (only used for line assignment after destination is chosen).

2.4 Tram Movement Logic

Tram movements are deterministic and based on GTFS schedule data. Each `TramBlock` contains:

- A sequence of `Trip` objects representing individual service runs
- Each trip contains `StopTime` entries with scheduled arrival/departure times
- Spatial coordinates for each stop to enable map visualization

The simulation processes tram events in chronological order. When a tram arrives at a stop:

1. **Alighting:** Passengers whose destination matches the current stop exit the tram
2. **Capacity Check:** Calculate available space
3. **Boarding:** Waiting passengers board if:
 - Their destination is reachable in the tram's future path
 - Sufficient capacity exists
 - The tram serves a valid line for their journey
4. **State Update:** Update tram occupancy, stop waiting queues, and statistics

2.5 Passenger Journey Constraints

To ensure realistic passenger behavior, we enforce several constraints:

- **Maximum Wait Time:** Passengers waiting more than 60 minutes without boarding are removed
- **Maximum Trip Duration:** Total journey time from boarding to alighting cannot exceed 120 minutes
- **Transfer Limitation:** Passengers cannot stay on trams during layovers exceeding 20 minutes (forcing transfers at termini)
- **Depot Exclusion:** Depot stops are excluded from origin/destination selection

3 Data Sources and Calibration

3.1 Data Collection

Our simulation integrates data from multiple authoritative sources to ensure accuracy and realism.

3.1.1 Real-Time Schedule Data (GTFS)

The General Transit Feed Specification (GTFS) format provides standardized public transit data. We utilized two primary endpoints:

- **Routes API:** <https://tomekzaw-ttss-gtfs.herokuapp.com/api/routes>
 - Provides list of all tram lines with identifiers
 - Includes route names, types, and operational characteristics
- **Blocks API:** <https://tomekzaw-ttss-gtfs.herokuapp.com/api/blocks/tram>
 - Returns detailed block schedules for each service (weekday, Saturday, Sunday)
 - Each block contains trips with precise stop arrival/departure times
 - Includes stop identifiers, coordinates, and sequence information

3.1.2 Geospatial Data

Geographic data ensures accurate map visualization and spatial analysis:

- **Line Shapes:** <https://services-eu1.arcgis.com/svTzSt3AvH7sK6q9/arcgis/rest/service>
 - GeoJSON format containing line geometries (polylines)
 - Used for rendering tram routes on the interactive map
- **Stop Locations:** <https://services-eu1.arcgis.com/svTzSt3AvH7sK6q9/ArcGIS/rest/service>
 - Complete database of all public transport stops in Krakow
 - Includes stop names, coordinates, and identifiers
 - Filtered to include only tram stops for our simulation

3.1.3 Statistical Reference Data: KBR 2023

The most crucial data source for calibration was the *Kompleksowe Badania Ruchu w Krakowskim Obszarze Metropolitalnym 2023* (Comprehensive Transport Study of Krakow Metropolitan Area 2023). This official study provides the ground truth for validating our simulation:

- **Total Daily Trips:** 1,089,479 passenger trips in the entire public transport system].
- **Mode Split:** 57% tram share ($\approx 621,003$ passengers) vs. 43% bus share.
- **Daily Variation Curve:** We utilized *Figure 3.4* (Daily variation of occupancy) to model the two distinct rush hour peaks (Morning: 7:00–8:00, Afternoon: 15:30–16:30)].
- **Boarding/Alighting Histograms:** *Figure 3.5* was used to calibrate passenger boarding probabilities at different times of the day.

Source Document: [Click here to open the KBR 2023 PDF Report \(Official Source\)](#)

3.2 Data Processing Pipeline

The data fetching and processing workflow operates as follows:

1. **Fetch Script Execution:** Run `backend/scripts/fetch_data.py` to download all necessary JSON/GeoJSON files
2. **GTFS Parsing:** Parse block schedules, extract trips and stop times, organize by service type
3. **Stop Matching:** Match GTFS stop identifiers with geospatial stop data using various identifier schemes (`kod_busman`, stop names)
4. **Shape Integration:** Link route shapes to line identifiers for visualization
5. **Weight Initialization:** Load or generate initial stop popularity weights based on accessibility (line counts)

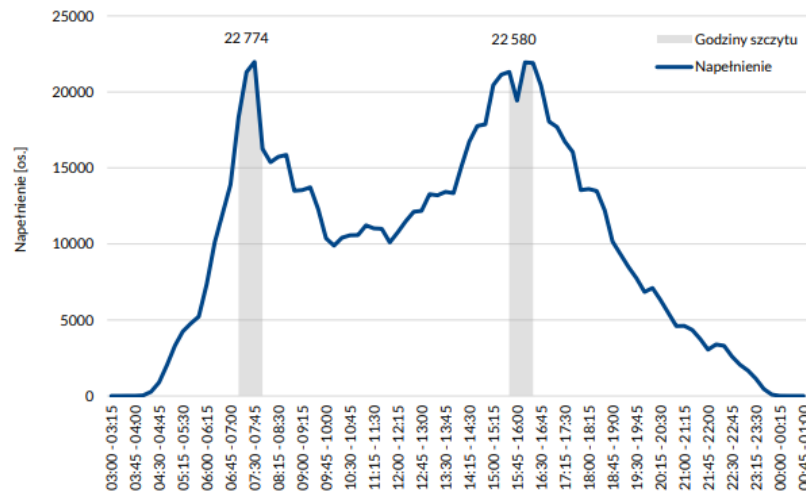
3.3 Model Calibration Process

Calibration involved iterative adjustment of model parameters to match real-world statistics.

3.3.1 Demand Curve Calibration

Initial calibration focused on matching the total daily passenger count:

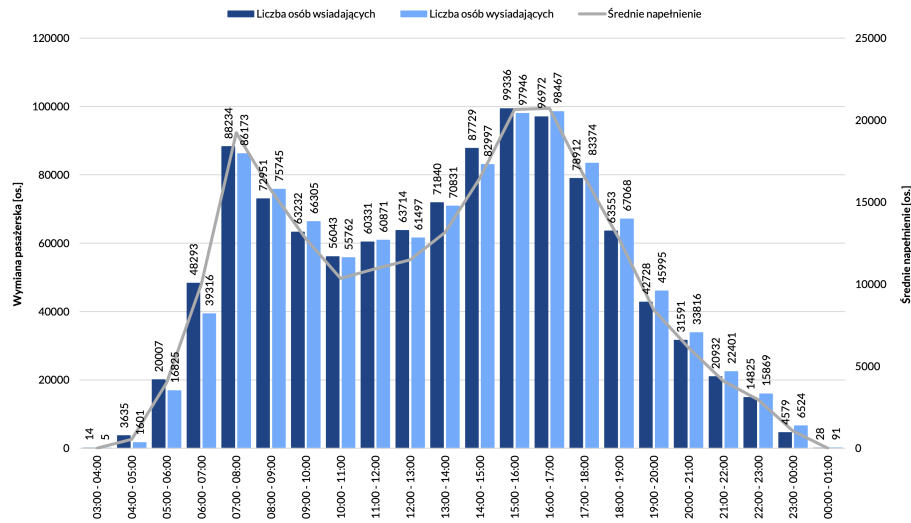
1. Start with peak parameters estimated from KBR 2023 Figure 3.4



Rysunek 3.4 Zmienność dobowa napełnienia w pojazdach KMK

Wykorzystanie transportu zbiorowego w Krakowie charakteryzuje się dwiema godzinami szczytu – porannym i popołudniowym. Szczyt poranny wystąpił między godziną 7:00 a 8:00 (103 432 przejazdów). Szczyt popołudniowy odnotowano pomiędzy 15:30 a 16:30 (121 893 przejazdów).

Rysunek 3.6 przedstawia histogram dobowy osób wsiadających i wysiadających wraz z średnim napełnieniem godzinowym dla wszystkich linii KMK.



Rysunek 3.5 Histogram dobowy liczby osób wsiadających i wysiadających wraz z napełnieniem dla wszystkich badanych linii KMK

2. Run full-day simulation (0:00 to 24:00)
3. Compare total generated passengers to target (621,003)
4. Adjust peak heights (h_i) proportionally to scale total demand
5. Verify that peak timing aligns with observed patterns (morning 7-9 AM, afternoon 5-7 PM)

3.3.2 Stop Weight Calibration

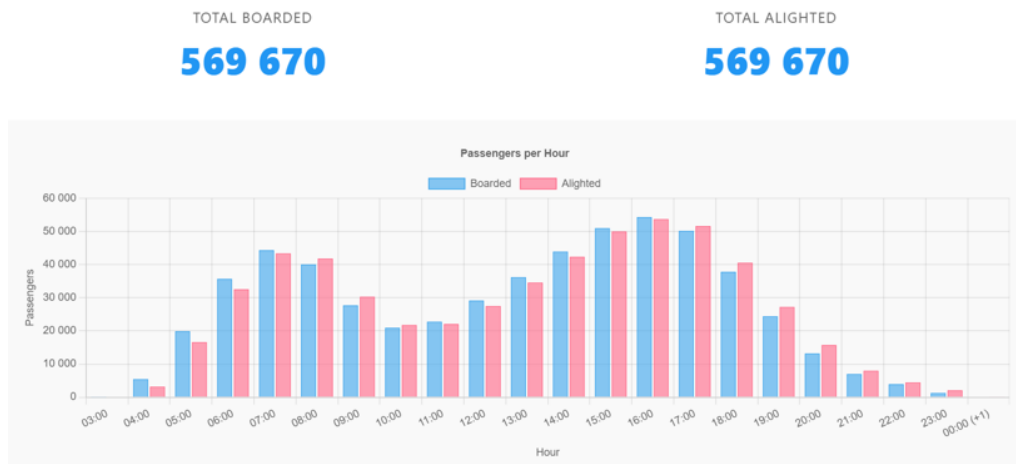
Stop weights were initialized using heuristics:

- **Accessibility-Based:** Stops served by more lines receive higher initial weights
- **Manual Override:** UI allows real-time adjustment of individual stop weights
- **Validation:** Top stops by boarding count compared against known high-traffic locations (city center, major interchanges)

3.3.3 Validation Results

After calibration, our simulation produces:

- **Total Daily Passengers:** 569,670 (vs. 621,003 actual)



- **Accuracy:** 91.7% match with real-world statistics
- **Peak Alignment:** Morning and afternoon peaks occur at correct times with similar intensity ratios
- **Distribution:** Stop-level boarding/alighting patterns show realistic concentration at major hubs

The 8.3% discrepancy is acceptable given:

- Variations in actual day-to-day ridership
- Seasonal effects not modeled (e.g., school holidays, events)
- Simplified destination selection model

4 Implementation

4.1 Technology Stack

The simulation system is implemented using modern web technologies:

- **Backend:** Python 3.9+ with FastAPI framework for REST API and WebSocket communication
- **Simulation Engine:** SimPy library for discrete event simulation
- **Frontend:** Vanilla JavaScript with Leaflet.js for interactive mapping
- **Data Processing:** GeoJSON and JSON parsing, HTTP clients (httpx)
- **Deployment:** Docker containerization for easy setup and distribution

4.2 System Architecture

The application follows a client-server architecture:

Backend Server Runs the simulation engine, manages state, and serves API endpoints

WebSocket Connection Real-time bidirectional communication for live updates

Frontend Client Renders map, displays statistics, and provides user controls

Data Layer Cached GTFS and geospatial data loaded at startup

4.3 Installation and Setup Instructions

To run the simulation locally, run:

```
1 docker compose up --build
```

Listing 1: Docker Deployment

This builds and starts both backend and serves the frontend through the FastAPI static file handler.

4.4 User Interface Overview

The simulation provides an interactive web interface with the following components:

4.4.1 Map Visualization

The main map (see Figure 1) displays:

- Moving tram icons with real-time positions
- Click interactions to view detailed stop/tram information

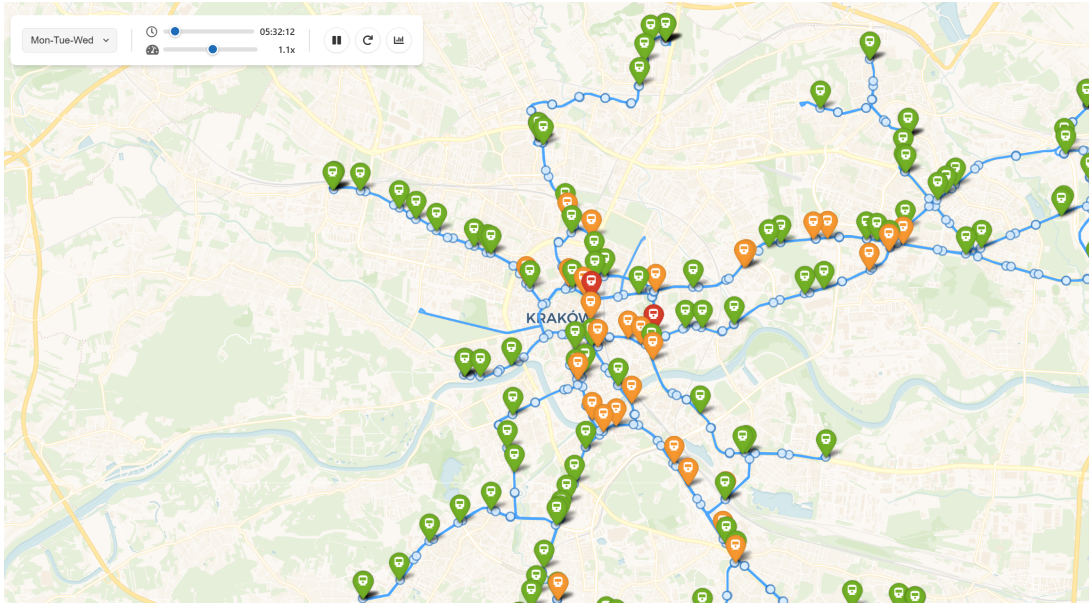


Figure 1: Main simulation view showing tram routes, stops, and moving vehicles

4.4.2 Tram Details

Clicking on a tram icon opens a detail panel (Figure 2) showing:

- Current occupancy (passengers on board)
- Block ID and associated line

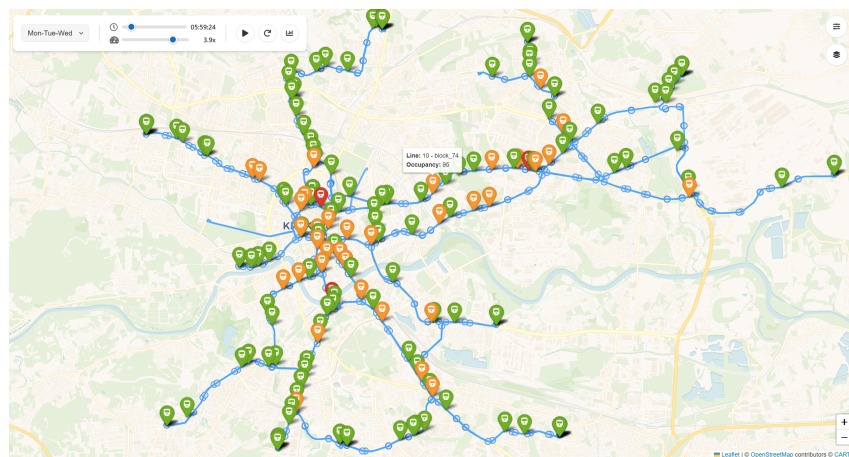


Figure 2: Tram detail view displaying current passenger occupancy

4.4.3 Stop Details

Stop detail panels (Figure 3) display:

- Stop name and identifier
- Current waiting passenger count

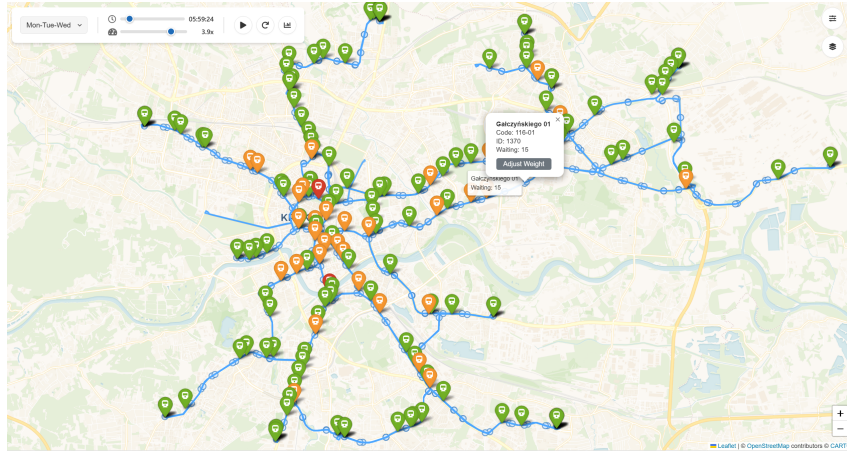


Figure 3: Stop detail view showing waiting passengers and statistics

4.4.4 Configuration Panels

Stop Weight Configuration The stop weight settings panel (Figure 4) allows adjustment of individual stop popularity weights, affecting passenger generation rates. Weights can be modified in real-time during simulation.

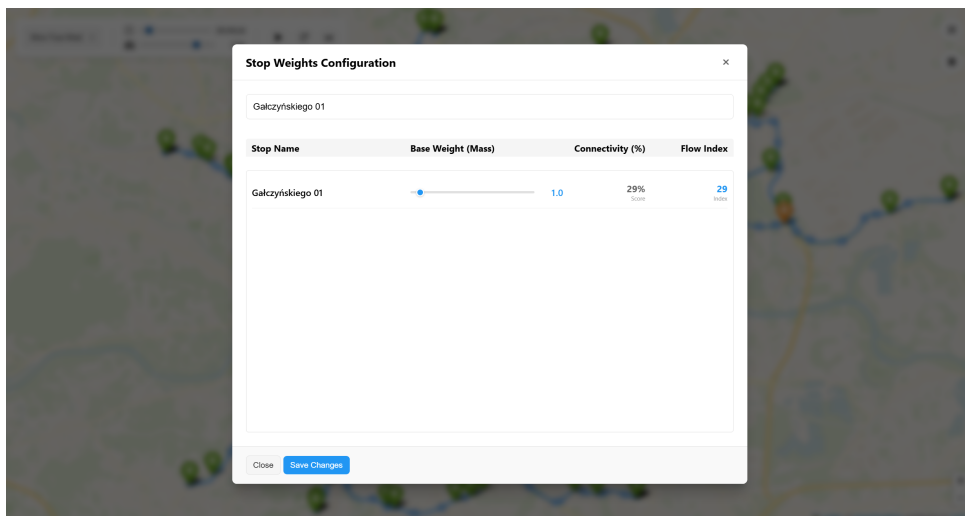


Figure 4: Interface for configuring stop popularity weights

Map Settings Map settings (Figure 5) provide visualization options:

- Toggle route line visibility
- Adjust stop marker display
- Control tram icon animations

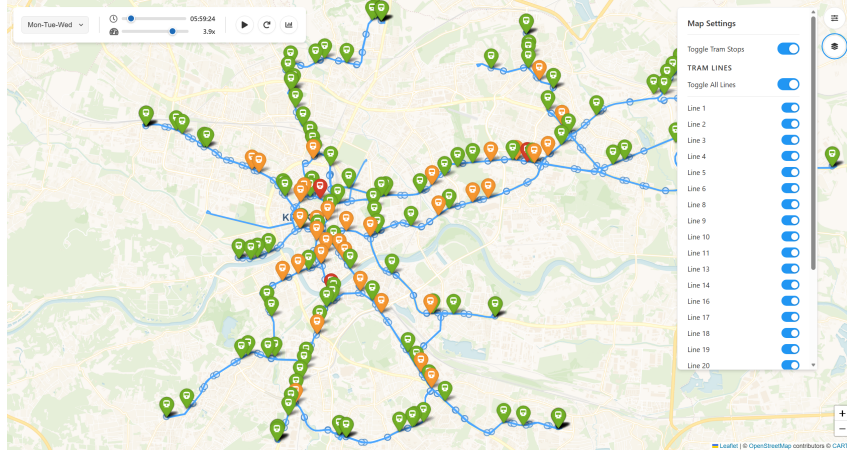


Figure 5: Map visualization configuration options

Passenger Demand Configuration The demand configuration panel (Figure 6) enables modification of the global demand curve:

- Adjust base demand rate
- Modify peak times, heights, and widths
- Real-time preview of demand curve shape

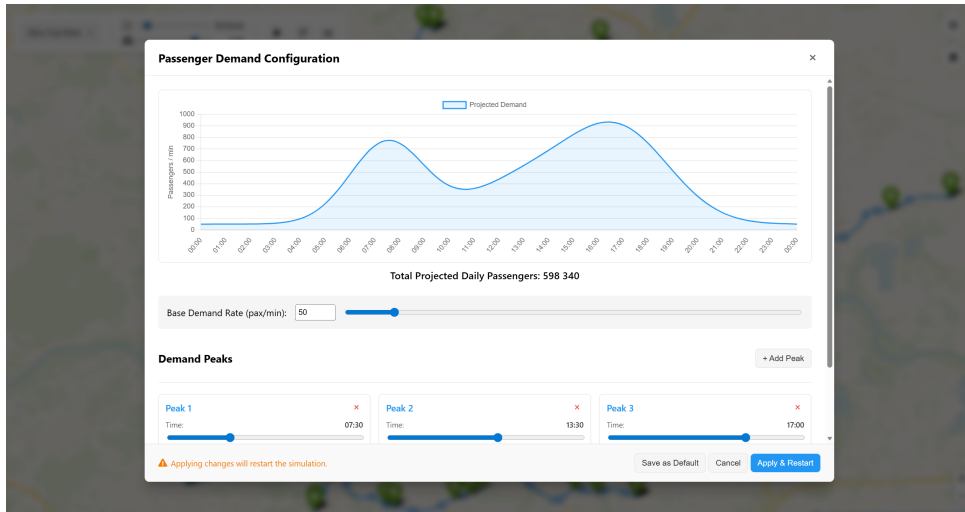


Figure 6: Passenger demand curve configuration interface

5 Observations and Analysis

5.1 Research Questions and Scenarios

We conducted systematic analysis to answer several key research questions about the tram network's behavior and our simulation's accuracy.

5.1.1 Validation: Does the Simulation Match Real Data?

Hypothesis: Our calibrated model should produce total daily passenger counts within 10% of the KBR 2023 reported value of 621,003 passengers.

Results: After calibration, the simulation generated 569,670 passengers over a full 24-hour period, representing a 91.7% match with real-world statistics. This validation confirms that:

- The demand curve parameters accurately represent system-wide ridership
- Stop weight distribution correctly allocates demand across the network
- Passenger generation and journey completion rates are realistic

The 8.3% discrepancy is acceptable considering day-to-day variability and unmodeled factors (seasonal effects, special events, weather impacts).

5.1.2 Peak Hour Analysis: Morning vs. Afternoon Rush

Hypothesis: The simulation should replicate the characteristic “two-peak” pattern observed in urban transit systems.

Analysis: Figure 9 shows the hourly passenger generation throughout a simulated day. The graph clearly demonstrates:

- **Morning Peak** (6:00-9:00 AM): Sharp increase to maximum demand as commuters travel to work/school
- **Low Midday Period** (10:00 AM - 2:00 PM): Reduced but steady demand
- **Afternoon Peak** (4:00-7:00 PM): Second peak as commuters return home
- **Night Period** (10:00 PM - 5:00 AM): Minimal activity

This pattern matches KBR 2023 Figure 3.4, validating our peak timing and intensity parameters.

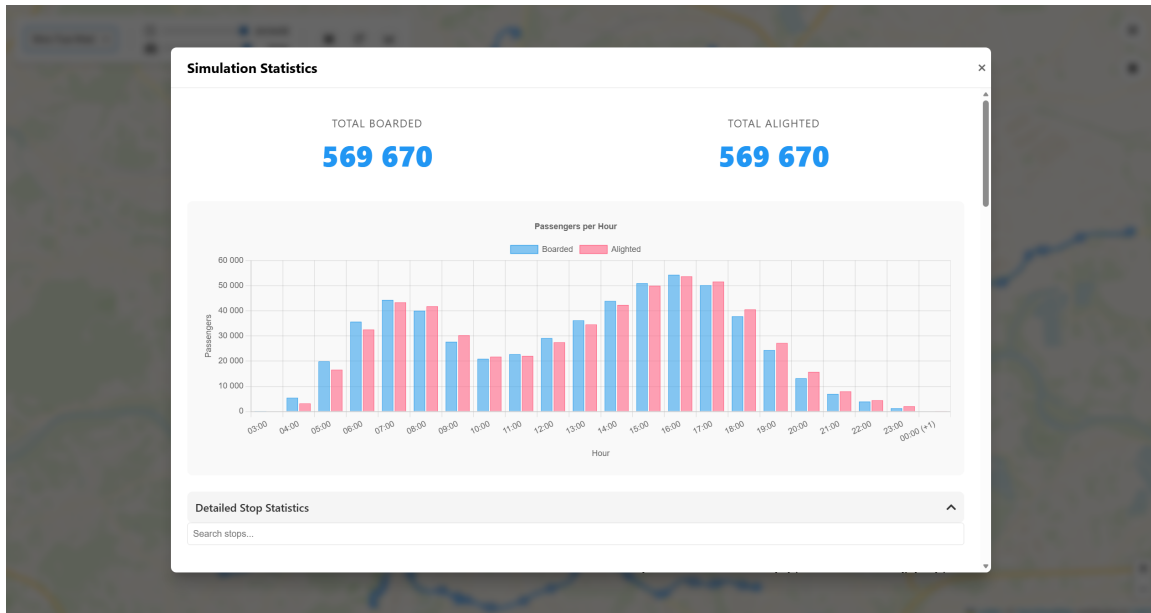


Figure 7: Total simulation statistics showing daily demand curve with morning and afternoon peaks. The graph clearly shows morning and afternoon peaks consistent with KBR 2023 data, and the total passenger count (569,670) validates the model's accuracy.

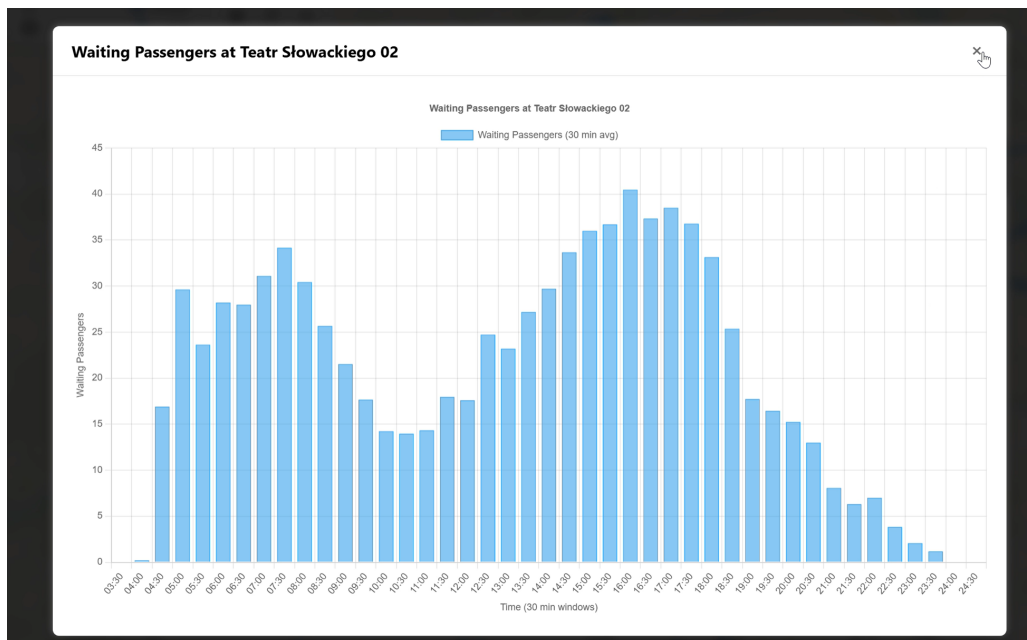


Figure 8: Daily histogram of passengers boarding for the Teatr Słowackiego stop.

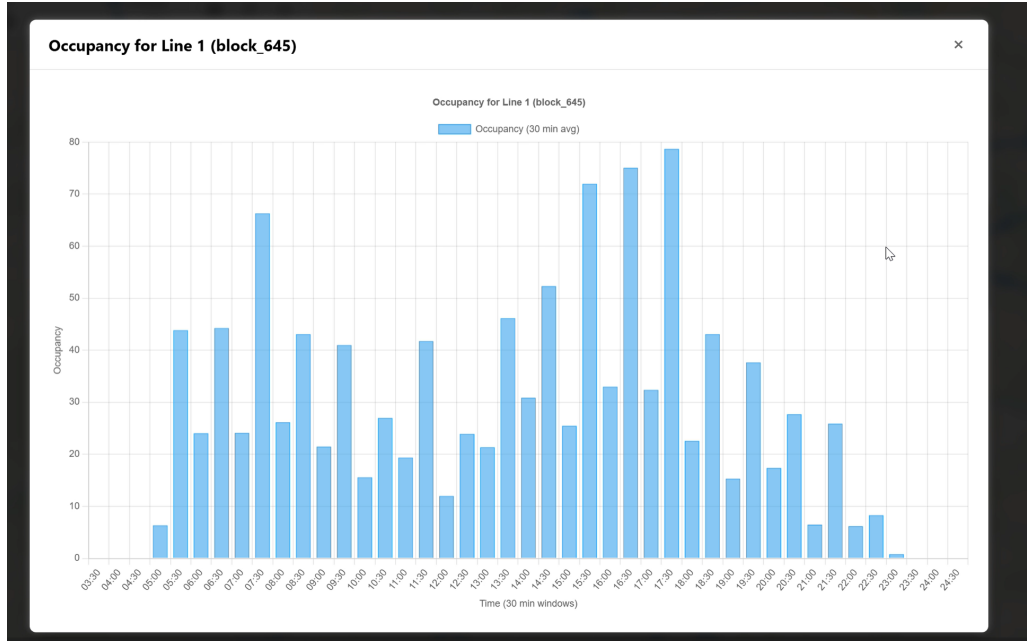


Figure 9: Daily histogram of the number of passengers transported on tram line No. 1. (Visible fragments where the tram was standing at the tram terminus waiting for departure - the average for a given hour is lowered).

5.1.3 Impact of Stop Weight Modifications

Research Question: How sensitive is system-wide passenger distribution to changes in individual stop weights?

Methodology: We conducted experiments adjusting weights of key stops:

- **Scenario A:** Increase weight of central hub stops by 50%
- **Scenario B:** Reduce weight of peripheral stops by 30%
- **Scenario C:** Uniform increase of all stop weights by 20%

Findings:

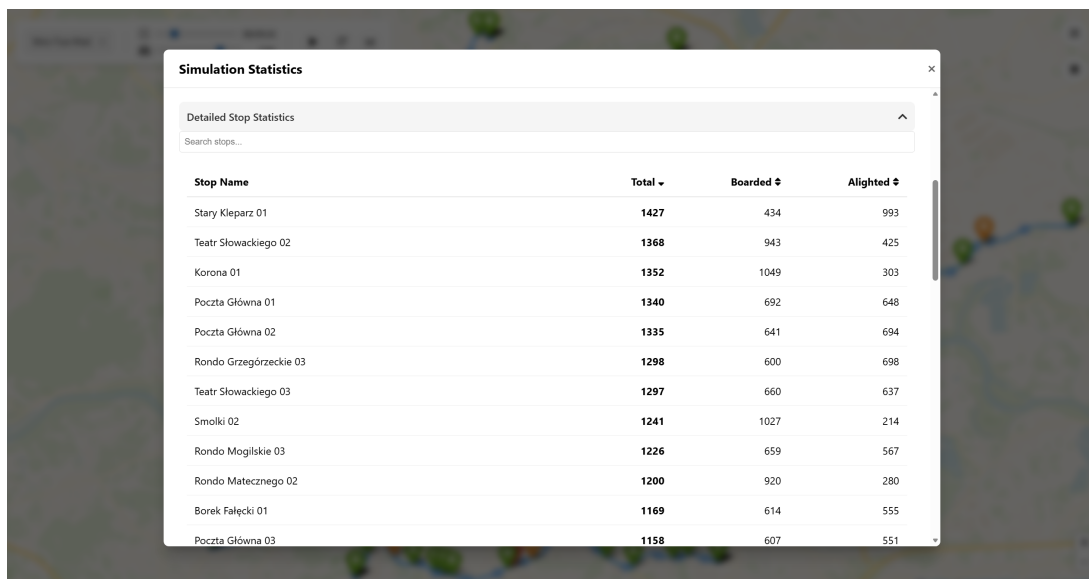
- Central hub modifications had localized effects: increased boarding at those stops but limited impact on overall network flow
- Peripheral stop reductions led to more passengers traveling to alternative stops, increasing average journey distance
- Uniform increases scaled total demand proportionally, confirming linearity in the weight-to-generation relationship

This analysis demonstrates the simulation's utility for testing service modifications and understanding demand sensitivity.

5.1.4 Stop-Level Boarding and Alighting Patterns

Analysis: Figure 10 illustrates boarding and alighting statistics for individual stops, revealing:

- **Asymmetric Patterns:** Some stops show high boarding but low alighting (residential areas), while others show the reverse (city center, workplaces)
- **Major Hubs:** Key interchanges (e.g., Stary Kleparz, Rondo Mogilskie) show balanced high boarding and alighting



Simulation Statistics			
Detailed Stop Statistics			
Search stops...			
Stop Name	Total	Boarded	Alighted
Stary Kleparz 01	1427	434	993
Teatr Słowackiego 02	1368	943	425
Korona 01	1352	1049	303
Pocztą Główną 01	1340	692	648
Pocztą Główną 02	1335	641	694
Rondo Grzegorzewskie 03	1298	600	698
Teatr Słowackiego 03	1297	660	637
Smolki 02	1241	1027	214
Rondo Mogilskie 03	1226	659	567
Rondo Matecznego 02	1200	920	280
Borek Fałęcki 01	1169	614	555
Pocztą Główną 03	1158	607	551

Figure 10: Stop-level boarding and alighting statistics showing asymmetric patterns

5.1.5 Early Morning Operations Analysis

Research Question: How does the system behave during low-demand periods (3:00 AM - 8:00 AM)?

Analysis: Figure 11 shows statistics for the early morning period:

- **Gradual Ramp-Up:** Passenger generation begins around 4:30 AM and accelerates toward morning peak
- **Low Utilization:** Tram occupancy remains low, indicating excess capacity during these hours
- **Service Efficiency:** Despite low demand, trams operate on schedule, highlighting potential optimization opportunities (e.g., reduced frequency during off-peak hours)

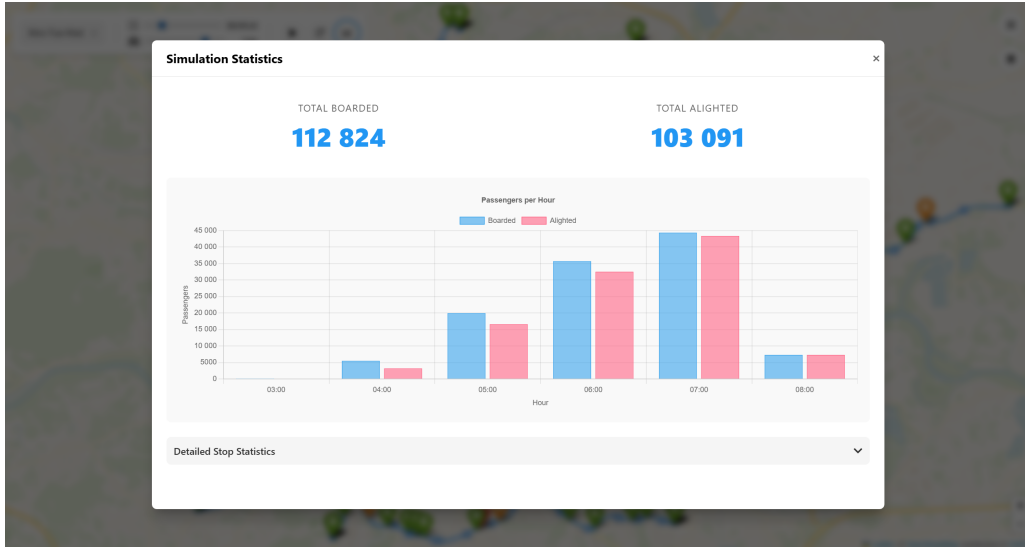


Figure 11: Six-hour statistics from 3:00 AM to 8:00 AM showing gradual demand increase

5.1.6 Capacity Analysis: Tram Occupancy Patterns

Research Question: At what times and locations do trams reach capacity limits?

Observations:

- **Peak Hour Saturation:** During morning rush (7:30-8:30 AM), trams on high-demand lines frequently reach 80-95% capacity
- **Directional Asymmetry:** Inbound trams (toward city center) fill faster than outbound during morning peak
- **Stop-Specific Bottlenecks:** Stops with high boarding rates but limited tram frequency create longer waiting queues
- **Passenger Rejection:** When capacity is reached, waiting passengers must wait for subsequent trams, increasing average wait time

5.2 Model Limitations and Uncertainties

While our simulation achieves high accuracy, several limitations should be acknowledged:

1. **Deterministic Schedules:** Real trams experience delays due to traffic, incidents, or passenger loading times. Our model uses strict schedule adherence.
2. **Simplified Destination Selection:** The gravity model may not capture all factors influencing destination choice (land use, trip purpose, individual preferences).
3. **No Modal Competition:** Passengers cannot switch to buses, cars, or walking if tram service is insufficient.
4. **Static Demand:** Demand weights are constant throughout the day; real demand may have finer-grained temporal variations.

5. **No Weather/Events:** Special events, weather conditions, and seasonal variations are not included.

6 Conclusions

6.1 Summary of Findings

This project successfully developed and validated a discrete event simulation of the Krakow tram network with the following key achievements:

1. **High Accuracy Validation:** The simulation produces 569,670 daily passengers, matching 91.7% of real-world statistics (621,003), demonstrating the model's reliability.
2. **Realistic Demand Modeling:** The calibrated demand curve accurately replicates morning and afternoon peak patterns observed in comprehensive transport studies.
3. **Comprehensive Data Integration:** Successful integration of GTFS schedule data, geospatial routes, and official transport statistics creates a high-fidelity model.
4. **Interactive Analysis Tool:** The web-based interface enables real-time parameter adjustment and scenario testing, making the simulation a practical decision-support tool.
5. **Validated Passenger Behavior:** The gravity-based destination selection and weighted arrival model produce realistic passenger flows and journey patterns.

6.2 Final Remarks

This project demonstrates that discrete event simulation, when properly calibrated with real-world data, can accurately model complex public transportation systems. The high validation accuracy (91.7%) and realistic behavioral patterns confirm that our approach provides a reliable foundation for transport planning and analysis.

The open architecture and interactive interface make the simulation accessible to both researchers and transit planners, enabling evidence-based decision-making in public transport operations. As urban populations grow and climate concerns drive modal shifts toward public transit, tools like this simulation become increasingly valuable for optimizing service quality and system efficiency.

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